

Lecture 10

Systems of ODE's

stability of initial-value problems for systems
numerical integration methods for systems
stability of numerical methods for systems
local and global truncation errors for systems

systems of ordinary differential equations

Systems of ODE's in vector notation

Let there be m unknown functions $y_j(t)$ satisfying m initial value problems:

$$\begin{aligned}y_j' &= f_j(t, y_1, \dots, y_m), \quad t > t_0; \\ y_j(t_0) &= y_{j,0}; \\ j &= 1, \dots, m.\end{aligned}$$

Introducing the vectors $\mathbf{y}, \mathbf{f} \in \mathbb{R}^m$:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}, \quad \mathbf{y}_0 = \begin{bmatrix} y_{1,0} \\ y_{2,0} \\ \vdots \\ y_{m,0} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1(t, y_1, \dots, y_m) \\ f_2(t, y_1, \dots, y_m) \\ \vdots \\ f_m(t, y_1, \dots, y_m) \end{bmatrix} = \begin{bmatrix} f_1(t, \mathbf{y}) \\ f_2(t, \mathbf{y}) \\ \vdots \\ f_m(t, \mathbf{y}) \end{bmatrix}$$

The system of ODE's can be written as

$$\begin{aligned}\mathbf{y}' &= \mathbf{f}(t, \mathbf{y}), \quad t \geq t_0; \\ \mathbf{y}(t_0) &= \mathbf{y}_0.\end{aligned}$$

On paper you write $\vec{y}$, $\vec{f}(t, \vec{y})$ or $\underline{y}$, $\underline{f}(t, \underline{y})$ when you work with vectors.

Systems of ODE's arising from higher-order ODE's

Consider an initial-value problem for the second-order ODE:

$$\begin{aligned}\phi'' &= f(t, \phi, \phi'), \quad t \geq t_0; \\ \phi(t_0) &= \phi_0, \\ \phi'(t_0) &= \phi'_0.\end{aligned}$$

Instead of looking for the unknown function $\phi(t)$ we can look for the pair of unknown functions $y_1(t) = \phi(t)$ and $y_2(t) = \phi'(t)$, putting them in a single vector as:

$$\mathbf{y}(t) = \begin{bmatrix} \phi(t) \\ \phi'(t) \end{bmatrix} = \begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}, \quad \mathbf{y}(t_0) = \begin{bmatrix} \phi_0 \\ \phi'_0 \end{bmatrix} = \begin{bmatrix} y_1(t_0) \\ y_2(t_0) \end{bmatrix} = \mathbf{y}_0$$

Systems of ODE's arising from higher-order ODE's

Taking the derivative of this vector we get the system of two first-order ODE's:

$$\begin{aligned}\mathbf{y}' &= \frac{d}{dt} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \frac{d}{dt} \begin{bmatrix} \phi \\ \phi' \end{bmatrix} = \begin{bmatrix} \phi' \\ \phi'' \end{bmatrix} = \begin{bmatrix} \phi' \\ f(t, \phi, \phi') \end{bmatrix} \\ &= \begin{bmatrix} y_2 \\ f(t, y_1, y_2) \end{bmatrix} = \begin{bmatrix} f_1(t, y_1, y_2) \\ f_2(t, y_1, y_2) \end{bmatrix} = \begin{bmatrix} f_1(t, \mathbf{y}) \\ f_2(t, \mathbf{y}) \end{bmatrix} = \mathbf{f}(t, \mathbf{y})\end{aligned}$$

Hence, in the componentwise notation this system of two ODE's can be written as:

$$\frac{d}{dt} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} y_2 \\ f(t, y_1, y_2) \end{bmatrix}, \quad \begin{bmatrix} y_1(t_0) \\ y_2(t_0) \end{bmatrix} = \begin{bmatrix} \phi_0 \\ \phi'_0 \end{bmatrix}$$

The shorthand (vector) notation is:

$$\mathbf{y}' = \mathbf{f}(t, \mathbf{y}), \quad \mathbf{y}(t_0) = \mathbf{y}_0$$

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Example: mathematical pendulum

Dynamics of an idealized pendulum is described by the initial-value problem for the second-order ODE:

$$\begin{aligned}\psi'' + \sin(\psi) &= 0, \quad t \geq 0; \\ \psi(0) &= \psi_0, \\ \psi'(0) &= 0.\end{aligned}$$

Introducing new unknown functions $y_1(t) = \psi(t)$ and $y_2(t) = \psi'(t)$, we rewrite this problem as an initial-value problem for the system of first-order ODE's:

$$\begin{aligned}\mathbf{y}' &= \mathbf{f}(t, \mathbf{y}), \quad t \geq 0; \\ \mathbf{y}(0) &= \mathbf{y}_0,\end{aligned}$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad \mathbf{y}_0 = \begin{bmatrix} \psi_0 \\ 0 \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} y_2 \\ -\sin(y_1) \end{bmatrix}$$

Example: linear second-order problem

A special case is the initial-value problem for a linear second-order ODE:

$$ax'' + bx' + cx = g, \quad t \geq 0;$$

$$x(0) = x_0,$$

$$x'(0) = x'_0.$$

Introducing new unknown functions $y_1(t) = x(t)$ and $y_2(t) = x'(t)$, we rewrite this problem as an initial-value problem for the system of first-order ODE's. To construct the system, proceed as follows:

$$y'_1 = y_2,$$

$$ay'_2 + by_2 + cy_1 = g, \Leftrightarrow y'_2 = -\frac{c}{a}y_1 - \frac{b}{a}y_2 + \frac{1}{a}g$$

$$\begin{aligned} \mathbf{y}' &= \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}' = \begin{bmatrix} y_2 \\ -\frac{c}{a}y_1 - \frac{b}{a}y_2 + \frac{1}{a}g \end{bmatrix} = \begin{bmatrix} (0)y_1 + (1)y_2 \\ -\frac{c}{a}y_1 - \frac{b}{a}y_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{a}g \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{a}g \end{bmatrix} = \mathbf{A}\mathbf{y} + \mathbf{g}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ \frac{1}{a}g \end{bmatrix}. \end{aligned}$$

stability of the test problem

Test problem and its perturbation

The test problem is:

$$\begin{aligned}\mathbf{y}' &= A\mathbf{y} + \mathbf{g}, \quad t \geq 0; \\ \mathbf{y}(0) &= \mathbf{y}_0,\end{aligned}$$

where the matrix $A \in \mathbb{R}^{m \times m}$ is constant in time. Perturbing the initial condition we get:

$$\begin{aligned}\tilde{\mathbf{y}}' &= A\tilde{\mathbf{y}} + \mathbf{g}, \quad t \geq 0; \\ \tilde{\mathbf{y}}(0) &= \mathbf{y}_0 + \epsilon_0.\end{aligned}$$

Let $\epsilon(t) = \tilde{\mathbf{y}}(t) - \mathbf{y}(t)$ be the perturbation. Then, it satisfies the problem

$$\begin{aligned}\epsilon' &= A\epsilon, \quad t \geq 0; \\ \epsilon(0) &= \epsilon_0.\end{aligned}$$

Diagonalizable system matrix

$$\begin{aligned}\epsilon' &= A\epsilon, \quad t \geq 0; \\ \epsilon(0) &= \epsilon_0.\end{aligned}$$

Assume that the matrix A is *diagonalizable*, i.e., that there exists an invertible matrix V such that:

$$A = V\Lambda V^{-1}, \quad V^{-1}AV = \Lambda.$$

The matrix $V \in \mathbb{R}^{m \times m}$ has the eigenvectors $\mathbf{v}_j$ as columns:

$$V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_m]$$

and the diagonal matrix $\Lambda \in \mathbb{R}^{m \times m}$ has the corresponding eigenvalues λ_j as its diagonal elements:

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & & & \\ 0 & \dots & & \lambda_m \end{bmatrix}$$

The eigenvector-eigenvalue relation is $A\mathbf{v}_j = \lambda_j\mathbf{v}_j$.

Solution with diagonalizable system matrix

To solve

$$\begin{aligned}\epsilon' &= A\epsilon, \quad t \geq 0; \\ \epsilon(0) &= \epsilon_0,\end{aligned}$$

introduce the representation of $\epsilon(t)$ via the eigenvectors of A as:

$$\epsilon(t) = V\eta(t)$$

Substituting we get:

$$\begin{aligned}V\eta' &= AV\eta, \quad t \geq 0; \\ V\eta_0 &= \epsilon_0.\end{aligned}$$

Multiplying by V^{-1} on the left we arrive at:

$$\begin{aligned}\eta' &= \Lambda\eta, \quad t \geq 0; \\ \eta_0 &= V^{-1}\epsilon_0.\end{aligned}$$

Solution with diagonalizable system matrix

The system

$$\begin{aligned}\eta' &= \Lambda \eta, \quad t \geq 0; \\ \eta_0 &= V^{-1} \epsilon_0.\end{aligned}$$

is *uncoupled*, i.e. all ODE's:

$$\begin{aligned}\eta_1' &= \lambda_1 \eta_1, \quad \eta_1(0) = \eta_{1,0}, \\ &\vdots \\ \eta_m' &= \lambda_m \eta_m, \quad \eta_m(0) = \eta_{m,0},\end{aligned}$$

can be solved individually as:

$$\begin{aligned}\eta_1(t) &= \eta_{1,0} e^{\lambda_1 t}, \\ &\vdots \\ \eta_m(t) &= \eta_{m,0} e^{\lambda_m t}.\end{aligned}$$

Solution with diagonalizable system matrix

In matrix notation this can be written as:

$$\boldsymbol{\eta}(t) = \exp(\boldsymbol{\Lambda}t)\boldsymbol{\eta}_0,$$

where

$$\exp(\boldsymbol{\Lambda}t) = \begin{bmatrix} \exp(\lambda_1 t) & 0 & \dots & 0 \\ 0 & \exp(\lambda_2 t) & & 0 \\ \vdots & & \ddots & \\ 0 & \dots & & \exp(\lambda_m t) \end{bmatrix}$$

Multiplying with V on the left and using $\boldsymbol{\eta}_0 = V^{-1}\boldsymbol{\epsilon}_0$ we obtain the solution:

$$\boldsymbol{\epsilon}(t) = V \exp(\boldsymbol{\Lambda}t) V^{-1} \boldsymbol{\epsilon}_0$$

Evolution of the perturbation

Solution

$$\epsilon(t) = V \exp(\Lambda t) V^{-1} \epsilon_0$$

can also be expressed as:

$$\epsilon(t) = \sum_{j=1}^m \alpha_j(t) \mathbf{v}_j,$$

where the coefficients $\alpha_j(t)$ have the form:

$$\alpha_j(t) = \beta_j e^{\lambda_j t},$$

with β_j being the j -th element of the vector $V^{-1} \epsilon_0$

If the matrix A has at least one eigenvalue λ_j with a positive real part:

$$\operatorname{Re}(\lambda_j) > 0,$$

then the coefficient $\alpha_j(t)$ will grow exponentially in time.

Definition: Stability of the test problem

The test problem

$$\begin{aligned} \mathbf{y}' &= A\mathbf{y} + \mathbf{g}, \quad t > 0; \\ \mathbf{y}(0) &= \mathbf{y}_0, \end{aligned}$$

with the diagonalizable matrix A is *stable* with respect to the perturbations in the initial condition, if the eigenvalues λ_j of A satisfy:

$$\operatorname{Re}(\lambda_j) \leq 0, \quad \text{for all } j = 1, \dots, m;$$

and *absolutely stable* if

$$\operatorname{Re}(\lambda_j) < 0, \quad \text{for all } j = 1, \dots, m.$$

If, for some j , $\operatorname{Re}(\lambda_j) > 0$, the problem is *unstable*.

Example: investigating stability of a problem

Consider the problem:

$$\begin{aligned} \mathbf{y}' &= A\mathbf{y}, \quad t > 0; \\ \mathbf{y}(0) &= \mathbf{y}_0, \end{aligned}$$

with the system matrix

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{bmatrix}$$

To study the stability, compute the eigenvalues of A , e.g., as roots of a polynomial equation:

$$\begin{aligned} \det(A - \lambda I) &= \det \left(\begin{bmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = \det \left(\begin{bmatrix} -\lambda & 1 \\ -\frac{c}{a} & -\frac{b}{a} - \lambda \end{bmatrix} \right) \\ &= (-\lambda) \left(-\frac{b}{a} - \lambda \right) - (1) \left(-\frac{c}{a} \right) = \lambda^2 + \frac{b}{a}\lambda + \frac{c}{a} = 0 \end{aligned}$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example: investigating stability of a problem

Since

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

stability of the problem depends on the values of a , b , and c .

a	b	c	λ_1	λ_2	stability
1	2	0	0	-2	stable
1	2	1	-1	-1	absolutely stable
1	0	1	$0 + i$	$0 - i$	stable
1	2	2	$-1 + i$	$-1 - i$	absolutely stable
1	-2	0	2	0	unstable

stability of general problems

From Lecture 4: stability of general scalar problems

For a general $f(t, y)$

$$\frac{dy}{dt} = f(t, y), \quad t > t_0; \quad y(0) = y_0,$$

consider the Taylor series around $\hat{t}$ and $\hat{y}$:

$$\begin{aligned} f(t, y) &= f(\hat{t}, \hat{y}) + \frac{\partial f(\hat{t}, \hat{y})}{\partial y}(y - \hat{y}) + \frac{\partial f(\hat{t}, \hat{y})}{\partial t}(t - \hat{t}) + C_1(t - \hat{t})^2 + C_2(y - \hat{y})^2 \\ &= \lambda y + g(t) + C_2(y - \hat{y})^2. \end{aligned}$$

Neglecting $C_2(y - \hat{y})^2$, this can be reduced to the test equation with $f(t, y) = \lambda y + g(t)$ and

$$\lambda = \frac{\partial f}{\partial y}(\hat{t}, \hat{y}).$$

Approximate (local) stability condition $\lambda < 0$, i.e., $\frac{\partial f}{\partial y}(\hat{t}, \hat{y}) < 0$, depends on $(\hat{t}, \hat{y})$.

Stability of general vector problems

For a general $\mathbf{f}(t, \mathbf{y})$

$$\mathbf{y}' = \mathbf{f}(t, \mathbf{y}), \quad t > t_0; \quad \mathbf{y}(0) = \mathbf{y}_0,$$

the Taylor series around $\hat{t}$ and $\hat{\mathbf{y}}$ looks like:

$$\mathbf{f}(t, \mathbf{y}) = J(\hat{t}, \hat{\mathbf{y}})\mathbf{y} + \mathbf{g}(t, \hat{t}, \hat{\mathbf{y}}) + \mathcal{O}(\|\mathbf{y} - \hat{\mathbf{y}}\|^2),$$

where the *Jacobian* matrix $J \in \mathbb{R}^{m \times m}$ is defined as:

$$J(\hat{t}, \hat{\mathbf{y}}) = \begin{bmatrix} \frac{\partial f_1}{\partial y_1} & \frac{\partial f_1}{\partial y_2} & \cdots & \frac{\partial f_1}{\partial y_m} \\ \frac{\partial f_2}{\partial y_1} & \frac{\partial f_2}{\partial y_2} & \cdots & \frac{\partial f_2}{\partial y_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial y_1} & \frac{\partial f_m}{\partial y_2} & \cdots & \frac{\partial f_m}{\partial y_m} \end{bmatrix} (\hat{t}, \hat{\mathbf{y}})$$

Definition: Local stability of a problem

The initial-value problem

$$\begin{aligned}\mathbf{y}' &= \mathbf{f}(t, \mathbf{y}), \quad t > 0; \\ \mathbf{y}(0) &= \mathbf{y}_0,\end{aligned}$$

is *stable* at $(\hat{t}, \hat{\mathbf{y}})$, if the eigenvalues λ_j of the Jacobian matrix $J(\hat{t}, \hat{\mathbf{y}})$ satisfy:

$$\operatorname{Re}(\lambda_j) \leq 0, \quad \text{for all } j = 1, \dots, m;$$

and *absolutely stable* at $(\hat{t}, \hat{\mathbf{y}})$ if

$$\operatorname{Re}(\lambda_j) < 0, \quad \text{for all } j = 1, \dots, m.$$

If, for some j , $\operatorname{Re}(\lambda_j) > 0$, the problem is *unstable* at $(\hat{t}, \hat{\mathbf{y}})$.

Stability properties of a dynamical system may change along its trajectory, since both J and its eigenvalues depend on $(\hat{t}, \hat{\mathbf{y}})$.

Typically, stability is investigated around *equilibrium* points, where $\mathbf{y}' = \mathbf{0}$, i.e., $\mathbf{f}(t, \mathbf{y}) = \mathbf{0}$.

Example: competing species

The population densities $y_1(t)$ and $y_2(t)$ of two competing species satisfy:

$$\begin{cases} y_1' &= y_1(1 - y_1 - y_2), \\ y_2' &= y_2(0.5 - 0.75y_1 - 0.25y_2), \end{cases} \quad t > t_0.$$

To study the stability of this nonlinear problem, compute the Jacobian matrix $J(y_1, y_2)$:

$$\begin{aligned} J(y_1, y_2) &= \begin{bmatrix} \frac{\partial}{\partial y_1}(y_1(1 - y_1 - y_2)) & \frac{\partial}{\partial y_2}(y_1(1 - y_1 - y_2)) \\ \frac{\partial}{\partial y_1}(y_2(0.5 - 0.75y_1 - 0.25y_2)) & \frac{\partial}{\partial y_2}(y_2(0.5 - 0.75y_1 - 0.25y_2)) \end{bmatrix} \\ &= \begin{bmatrix} 1 - 2y_1 - y_2 & -y_1 \\ -0.75y_2 & 0.5 - 0.75y_1 - 0.5y_2 \end{bmatrix} \end{aligned}$$

Example: competing species

Consider, for example, the point $(\hat{y}_1, \hat{y}_2) = (1.5, 0)$:

$$J(1.5, 0) = \begin{bmatrix} 1 - 2(1.5) - (0) & -1.5 \\ -0.75(0) & 0.5 - 0.75(1.5) - 0.5(0) \end{bmatrix} = \begin{bmatrix} -2 & -1.5 \\ 0 & -0.625 \end{bmatrix}$$

Compute the eigenvalues of $J(1.5, 0)$:

$$\det \left(\begin{bmatrix} -2 - \lambda & -1.5 \\ 0 & -0.625 - \lambda \end{bmatrix} \right) = (-2 - \lambda)(-0.625 - \lambda) = 0$$

The eigenvalues are:

$$\lambda_1 = -2, \quad \lambda_2 = -0.625$$

Since $\lambda_1 < 0$ and $\lambda_2 < 0$, the problem is *absolutely stable* at $(1.5, 0)$.

Summary: Stability of systems of ODE's

- Stability is always studied on the linear test system of the form $\epsilon' = A\epsilon$.
- For general systems $\mathbf{y}' = \mathbf{f}(t, \mathbf{y})$, stability is studied around some point $(\hat{t}, \hat{\mathbf{y}})$ on the linearized test system $\epsilon' = J(\hat{t}, \hat{\mathbf{y}})\epsilon$, where $J(\hat{t}, \hat{\mathbf{y}})$ is the Jacobian matrix at $(\hat{t}, \hat{\mathbf{y}})$.
- To study the stability, you need to calculate the eigenvalues of A or $J(\hat{t}, \hat{\mathbf{y}})$
- Stability of the system depends on the sign of the real part of the eigenvalues. The presence of just one eigenvalue with a positive real part means instability of the whole system.
- When stability is studied at some point $(\hat{t}, \hat{\mathbf{y}})$, one can determine the *type* of this point (stable node, stable/unstable spiral point, circular point, etc)

numerical integration of systems of ODE's

Numerical integration is applied componentwise

$$\mathbf{y}' = \mathbf{f}(t, \mathbf{y}),$$
$$\int_{t_n}^{t_{n+1}} \mathbf{y}' dt = \begin{bmatrix} \int_{t_n}^{t_{n+1}} y'_1 dt \\ \vdots \\ \int_{t_n}^{t_{n+1}} y'_m dt \end{bmatrix} = \begin{bmatrix} \int_{t_n}^{t_{n+1}} f_1(t, \mathbf{y}) dt \\ \vdots \\ \int_{t_n}^{t_{n+1}} f_m(t, \mathbf{y}) dt \end{bmatrix} = \int_{t_n}^{t_{n+1}} \mathbf{f}(t, \mathbf{y}) dt,$$

$$\mathbf{y}_{n+1} - \mathbf{y}_n = \int_{t_n}^{t_{n+1}} \mathbf{f}(t, \mathbf{y}) dt,$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \int_{t_n}^{t_{n+1}} \mathbf{f}(t, \mathbf{y}) dt,$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \sum_{k=1}^m \alpha_k \mathbf{f}(t_k, \mathbf{y}_k) + \mathcal{O}((\Delta t)^p),$$

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \sum_{k=1}^m \alpha_k \mathbf{f}(t_k, \mathbf{w}_k),$$

Common methods

Forward Euler

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \Delta t \mathbf{f}(t_n, \mathbf{w}_n)$$

Backward Euler

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \Delta t \mathbf{f}(t_{n+1}, \mathbf{w}_{n+1}), \quad \text{and solve for } \mathbf{w}_{n+1}$$

Trapezoidal

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \frac{\Delta t}{2} \mathbf{f}(t_n, \mathbf{w}_n) + \frac{\Delta t}{2} \mathbf{f}(t_{n+1}, \mathbf{w}_{n+1}), \quad \text{and solve for } \mathbf{w}_{n+1}$$

Modified Euler

$$\bar{\mathbf{w}}_{n+1} = \mathbf{w}_n + \Delta t \mathbf{f}(t_n, \mathbf{w}_n), \quad \text{predictor}$$

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \frac{\Delta t}{2} \mathbf{f}(t_n, \mathbf{w}_n) + \frac{\Delta t}{2} \mathbf{f}(t_{n+1}, \bar{\mathbf{w}}_{n+1}), \quad \text{corrector}$$

Runge-Kutta (RK4) method

$$\mathbf{k}_1 = \Delta t \mathbf{f}(t_n, \mathbf{w}_n),$$

$$\mathbf{k}_2 = \Delta t \mathbf{f}(t_n + \Delta t/2, \mathbf{w}_n + \mathbf{k}_1/2),$$

$$\mathbf{k}_3 = \Delta t \mathbf{f}(t_n + \Delta t/2, \mathbf{w}_n + \mathbf{k}_2/2),$$

$$\mathbf{k}_4 = \Delta t \mathbf{f}(t_n + \Delta t, \mathbf{w}_n + \mathbf{k}_3),$$

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).$$

Example: simulating mathematical pendulum with FE method

Applying the Forward Euler method to the initial-value problem

$$\begin{aligned} \mathbf{y}' &= \mathbf{f}(t, \mathbf{y}), \quad t \geq 0; \\ \mathbf{y}(0) &= \mathbf{y}_0, \end{aligned}$$

we get iterations

$$\begin{aligned} \mathbf{w}_0 &= \mathbf{y}_0, \\ \mathbf{w}_{n+1} &= \mathbf{w}_n + \Delta t \mathbf{f}(t, \mathbf{w}_n), \quad n = 0, 1, \dots \end{aligned}$$

For the mathematical pendulum with $\mathbf{f}(y_1, y_2) = [y_2, \sin(y_1)]^T$ this means:

$$\begin{aligned} w_{1,0} &= \psi_0, \quad w_{2,0} = 0, \\ \begin{bmatrix} w_{1,n+1} \\ w_{2,n+1} \end{bmatrix} &= \begin{bmatrix} w_{1,n} \\ w_{2,n} \end{bmatrix} + \Delta t \begin{bmatrix} w_{2,n} \\ \sin(w_{1,n}) \end{bmatrix}, \quad n = 0, 1, \dots \\ \begin{cases} w_{1,n+1} &= w_{1,n} + \Delta t w_{2,n} \\ w_{2,n+1} &= w_{2,n} + \Delta t \sin(w_{1,n}), \end{cases} \quad n = 0, 1, \dots \end{aligned}$$

Example: integrating linear systems with FE method

Linear second-order ODE's and many other lead to the systems of the kind:

$$\begin{aligned}\mathbf{y}' &= A\mathbf{y} + \mathbf{g}, \quad t \geq 0; \\ \mathbf{y}(0) &= \mathbf{y}_0.\end{aligned}$$

If the matrix $A \in \mathbb{R}^{m \times m}$ does not depend on time, then the Forward Euler method will produce the iterations:

$$\begin{aligned}\mathbf{w}_0 &= \mathbf{y}_0, \\ \mathbf{w}_{n+1} &= \mathbf{w}_n + \Delta t (A\mathbf{w}_n + \mathbf{g}_n) = (I + \Delta t A)\mathbf{w}_n + \Delta t \mathbf{g}_n, \quad n = 0, 1, \dots\end{aligned}$$

where I is the identity matrix.

Example: integrating linear systems with BE method

The Backward Euler method will produce iterations:

$$\begin{aligned}\mathbf{w}_0 &= \mathbf{y}_0, \\ \mathbf{w}_{n+1} &= \mathbf{w}_n + \Delta t (A\mathbf{w}_{n+1} + \mathbf{g}_{n+1}), \quad n = 0, 1, \dots\end{aligned}$$

This is equivalent to:

$$\begin{aligned}\mathbf{w}_0 &= \mathbf{y}_0, \\ [I - \Delta t A]\mathbf{w}_{n+1} &= \mathbf{w}_n + \Delta t \mathbf{g}_{n+1}, \quad n = 0, 1, \dots\end{aligned}$$

Hence, to find $\mathbf{w}_{n+1}$, a system of linear algebraic equations must be solved at each time step. If the matrix $[I - \Delta t A]$ is invertible, formally, we can write:

$$\begin{aligned}\mathbf{w}_0 &= \mathbf{y}_0, \\ \mathbf{w}_{n+1} &= [I - \Delta t A]^{-1}(\mathbf{w}_n + \Delta t \mathbf{g}_{n+1}), \quad n = 0, 1, \dots\end{aligned}$$

stability of numerical methods for systems of ODE's

Test equation for numerical methods

Stability of numerical methods is studied on the test equation:

$$\begin{aligned}\mathbf{y}' &= A\mathbf{y} + \mathbf{g}, \quad t > 0; \\ \mathbf{y}(0) &= \mathbf{y}_0,\end{aligned}$$

and its perturbed version:

$$\begin{aligned}\tilde{\mathbf{y}}' &= A\tilde{\mathbf{y}} + \mathbf{g}, \quad t > 0; \\ \tilde{\mathbf{y}}(0) &= \mathbf{y}_0 + \epsilon_0.\end{aligned}$$

The perturbation $\epsilon(t) = \tilde{\mathbf{y}}(t) - \mathbf{y}(t)$ obeys:

$$\begin{aligned}\epsilon' &= A\epsilon, \quad t > 0; \\ \epsilon(0) &= \epsilon_0,\end{aligned}$$

For general problems, stability is studied around some point $(\hat{t}, \hat{\mathbf{y}})$. In which case, $A = J(\hat{t}, \hat{\mathbf{y}})$.

Amplification matrix

To investigate the stability of a numerical method that produces approximations $\mathbf{w}_n$ for $\mathbf{y}_n$, the method should be applied to the problem:

$$\begin{aligned}\epsilon' &= A\epsilon, \quad t > 0; \\ \epsilon(0) &= \epsilon_0,\end{aligned}$$

and written in the form:

$$\mathbf{w}_{n+1} = Q\mathbf{w}_n, \quad n = 0, 1, \dots$$

where $Q \in \mathbb{R}^{m \times m}$ is the *amplification matrix* of the method.

Example: amplification matrix of the FE method

Applying the Forward Euler method to

$$\begin{aligned}\epsilon' &= A\epsilon, \quad t > 0; \\ \epsilon(0) &= \epsilon_0,\end{aligned}$$

we get:

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \Delta t A \mathbf{w}_n = [I + \Delta t A] \mathbf{w}_n.$$

Hence, the amplification matrix of the Forward Euler method is:

$$Q = I + \Delta t A$$

For implicit methods, invertibility of certain matrices must be assumed.

Amplification matrices of common numerical methods

Method	Amplification matrix Q
Forward Euler	$I + \Delta t A$
Backward Euler	$[I - \Delta t A]^{-1}$
Trapezoidal	$[I - \frac{1}{2}\Delta t A]^{-1} [I + \frac{1}{2}\Delta t A]$
Modified Euler	$I + \Delta t A + \frac{(\Delta t)^2}{2} A^2$
Runge-Kutte (RK4)	$I + \Delta t A + \frac{(\Delta t)^2}{2} A^2 + \frac{(\Delta t)^3}{6} A^3 + \frac{(\Delta t)^4}{24} A^4$

Amplification matrix and stability

Since

$$\mathbf{w}_{n+1} = Q\mathbf{w}_n = Q^n\mathbf{w}_0, \quad n = 0, 1, \dots$$

we intuitively expect a stability condition of the form " $|Q| \leq 1$ ", but this time for the matrix Q .

Diagonalizable amplification matrices

Assume that Q is diagonalizable, i.e., that there exists an invertible matrix V with the eigenvectors $\mathbf{v}_j$ of Q as columns, such that $Q = V\tilde{\Lambda}V^{-1}$, where $\tilde{\Lambda}$ is the diagonal matrix with the eigenvalues $\tilde{\lambda}_j$ of Q as its diagonal elements. Then (why?),

$$\mathbf{w}_{n+1} = Q^n \mathbf{w}_0 = (V\tilde{\Lambda}V^{-1})^n \mathbf{w}_0 = V\tilde{\Lambda}^n V^{-1} \mathbf{w}_0$$

This can also be written as:

$$\mathbf{w}_{n+1} = \sum_{j=0}^m \alpha_j \mathbf{v}_j = \sum_{j=0}^m b_j \tilde{\lambda}_j^n \mathbf{v}_j,$$

where b_j denotes the j -th element of the vector $V^{-1}\mathbf{w}_0$.

To keep $\mathbf{w}_{n+1}$ bounded as $n \rightarrow \infty$, the magnitudes of all α_j 's and therefore the magnitudes of all $\tilde{\lambda}_j^n$'s should be bounded.

Definition: Stability of a numerical method for a system of ODE's

Let a numerical method have the diagonalizable amplification matrix $Q \in \mathbb{R}^{m \times m}$ with the eigenvalues $\tilde{\lambda}_j, j = 1, \dots, m$. Then, the method is *stable* if:

$$|\tilde{\lambda}_j| \leq 1, \quad \text{for all } j = 1, \dots, m;$$

absolutely stable if

$$|\tilde{\lambda}_j| < 1, \quad \text{for all } j = 1, \dots, m;$$

and *unstable* if

$$|\tilde{\lambda}_j| > 1, \quad \text{for any } j = 1, \dots, m.$$

It is important to understand the difference between '*for all*' and '*for any*' here.

Example: Stability of the FE method

The amplification matrix of the Forward Euler method is:

$$Q = I + \Delta t A$$

Let λ_j be the eigenvalues of A . Then, the eigenvalues of Q are (why?)

$$\tilde{\lambda}_j = 1 + \Delta t \lambda_j$$

Let $\lambda_j = \lambda'_j + i\lambda''_j$, where $\lambda'_j = \text{Re}(\lambda_j)$ and $\lambda''_j = \text{Im}(\lambda_j)$. Stability condition becomes:

$$\begin{aligned} |\tilde{\lambda}_j| &= |1 + \Delta t \lambda_j| \leq 1, \\ |1 + \Delta t \lambda'_j + i\Delta t \lambda''_j|^2 &\leq 1^2, \\ (1 + \Delta t \lambda'_j)^2 + (\Delta t \lambda''_j)^2 &\leq 1, \\ 1 + 2\lambda'_j\Delta t + ((\lambda'_j)^2 + (\lambda''_j)^2)(\Delta t)^2 &\leq 1, \\ \Delta t (2\lambda'_j + |\lambda_j|^2\Delta t) &\leq 0, \quad \text{for all } j = 1, \dots, m. \end{aligned}$$

Example: Stability of the FE method

To satisfy the condition:

$$\Delta t (2\lambda'_j + |\lambda_j|^2 \Delta t) \leq 0, \quad \text{for all } j = 1, \dots, m;$$

assume that $\Delta t > 0$. Then,

$$2\lambda'_j + |\lambda_j|^2 \Delta t \leq 0, \quad \text{for all } j = 1, \dots, m;$$

Now, recall that the test problem is absolutely stable if $\text{Re}(\lambda_j) < 0$, i.e., $\lambda'_j = -|\lambda'_j|$, and $|\lambda'_j| > 0$. Assuming that the test problem is *absolutely stable*,

$$-2|\lambda'_j| + |\lambda_j|^2 \Delta t \leq 0, \quad \lambda'_j < 0;$$

$$|\lambda_j|^2 \Delta t \leq 2|\lambda'_j|,$$

$$\Delta t \leq \frac{2|\lambda'_j|}{|\lambda_j|^2} = \frac{2|\lambda'_j|}{(\lambda'_j)^2 + (\lambda''_j)^2}, \quad \text{for all } j = 1, \dots, m;$$

The Forward Euler method for stable systems of ODE's is only conditionally stable.

Example: Stability of the BE method

The amplification matrix of the Backward Euler method is:

$$Q = [I - \Delta t A]^{-1}$$

Let λ_j be the eigenvalues of A . Then, the eigenvalues of Q are (why?)

$$\tilde{\lambda}_j = \frac{1}{1 - \Delta t \lambda_j}$$

Let $\lambda_j = \lambda'_j + i\lambda''_j$, where $\lambda'_j = \text{Re}(\lambda_j)$ and $\lambda''_j = \text{Im}(\lambda_j)$. Stability condition becomes:

$$|\tilde{\lambda}_j| = \frac{1}{|1 - \Delta t \lambda_j|} \leq 1,$$

$$1 \leq |1 - \Delta t \lambda'_j - i\Delta t \lambda''_j|,$$

$$1 \leq |1 - \Delta t \lambda'_j - i\Delta t \lambda''_j|^2,$$

$$1 \leq (1 - \Delta t \lambda'_j)^2 + (\Delta t \lambda''_j)^2,$$

$$1 \leq 1 - 2\Delta t \lambda'_j + |\lambda_j|^2(\Delta t)^2,$$

$$2\Delta t \lambda'_j \leq |\lambda_j|^2(\Delta t)^2, \quad \text{for all } j = 1, \dots, m.$$

Example: Stability of the BE method

To satisfy the condition:

$$2\Delta t \lambda'_j \leq |\lambda_j|^2 (\Delta t)^2, \quad \text{for all } j = 1, \dots, m;$$

assume that $\Delta t > 0$. Then,

$$2\lambda'_j \leq |\lambda_j|^2 \Delta t, \quad \text{for all } j = 1, \dots, m;$$

For stable test problems with $\lambda'_j = -|\lambda_j| \leq 0$, the inequality

$$-2|\lambda'_j| \leq |\lambda_j|^2 \Delta t, \quad \text{for all } j = 1, \dots, m;$$

is satisfied for any $\Delta t > 0$, making the Backward Euler method unconditionally stable when applied to stable systems of ODE's.

Stability diagrams

On p. 90 of the book there is Figure 6.4 that shows the regions of stability for the common numerical methods as shaded area in the two-dimensional plane with

$$\operatorname{Re}(\lambda_j \Delta t) = \lambda'_j \Delta t, \quad \text{as } x \text{ coordinate,}$$

$$\operatorname{Im}(\lambda_j \Delta t) = \lambda''_j \Delta t, \quad \text{as } y \text{ coordinate.}$$

The method is stable if for every λ_j the chosen Δt corresponds to a point inside or on the boundary of the shaded area.

Purely imaginary eigenvalues

What happens if $\operatorname{Re}(\lambda_j) = \lambda'_j = 0$? The test problem is stable in that case and the implicit Backward Euler method is also stable.

However, recalling that the explicit Forward Euler method must satisfy the stability condition:

$$\begin{aligned} -2|\lambda'_j| + |\lambda_j|^2 \Delta t &\leq 0, \\ (\lambda''_j)^2 \Delta t &\leq 0, \end{aligned}$$

we arrive at the absurd choice $\Delta t = 0$. The same is true for the explicit Modified Euler method.

The only common *explicit* method that is stable with purely imaginary λ_j is the Runge-Kutta RK4 method. It is conditionally stable with the stability condition:

$$|\lambda_j| \Delta t \leq 2.8, \quad \text{for all } j = 1, \dots, m.$$

Summary: Stability of numerical methods for systems of ODE's

- Stability of numerical time-integration methods is also studied on the linear test system of the form $\epsilon' = A\epsilon$. For general systems $\mathbf{y}' = \mathbf{f}(t, \mathbf{y})$, stability is studied around some point $(\hat{t}, \hat{\mathbf{y}})$ on the linearized test system $\epsilon' = J(\hat{t}, \hat{\mathbf{y}})\epsilon$, where $J(\hat{t}, \hat{\mathbf{y}})$ is the Jacobian matrix at $(\hat{t}, \hat{\mathbf{y}})$.
- To study the stability of the numerical time-integration method:
 - 1 derive the amplification matrix Q of the method, which depends on A or J and Δt .
 - 2 derive or estimate the eigenvalues of Q . Usually, the eigenvalues $\tilde{\lambda}_j$ of Q can be expressed through the eigenvalues λ_j of A or J .
 - 3 the general stability condition $|\tilde{\lambda}_j| \leq 1$ for all $j = 1, \dots, m$; leads to the stability condition on Δt expressed as an inequality in terms of the eigenvalues λ_j of A or J .
- There are conditionally and unconditionally stable numerical methods. Explicit numerical methods do not always work with purely imaginary λ_j .

local and global truncation errors for systems

Definition: local truncation error for systems

Let $\mathbf{y}_n = \mathbf{y}(t_n)$ be given and $\mathbf{y}_{n+1} = \mathbf{y}(t_{n+1})$ be the exact solution of the initial-value problem at t_{n+1} and $\mathbf{z}_{n+1}$ its numerical approximation starting with the known $\mathbf{z}_n = \mathbf{y}_n$. Then, the *local truncation error* τ_{n+1} is defined as:

$$\tau_{n+1} = \frac{1}{\Delta t}(\mathbf{y}_{n+1} - \mathbf{z}_{n+1}),$$

where $\Delta t = t_{n+1} - t_n$.

Definition: global truncation error for systems

The *global truncation error* $\mathbf{e}_{n+1}$ is defined as the difference between the exact solution $\mathbf{y}_{n+1}$ at t_{n+1} and its numerical approximation $\mathbf{w}_{n+1}$:

$$\mathbf{e}_{n+1} = \mathbf{y}_{n+1} - \mathbf{w}_{n+1},$$

where $\mathbf{w}_0 = \mathbf{y}_0$.

Local truncation error of the FE method for systems

$$\begin{aligned} \mathbf{y}' &= \mathbf{f}(t, \mathbf{y}), \quad t_n \leq t \leq t_{n+1}, \\ \mathbf{y}(t_n) &= \mathbf{y}_n \end{aligned}$$

Forward Euler method proceeds as:

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \Delta t \mathbf{f}(t_n, \mathbf{w}_n), \quad n = 0, 1, \dots$$

Let $\mathbf{w}_n = \mathbf{y}_n$, then:

$$\mathbf{z}_{n+1} = \mathbf{y}_n + \Delta t \mathbf{f}(t_n, \mathbf{y}_n)$$

For the exact $\mathbf{y}_{n+1}$ we use the Taylor series (for each component of $\mathbf{y}_{n+1}$):

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \mathbf{y}'_n \Delta t + \frac{\mathbf{y}''(\xi)}{2} (\Delta t)^2 = \mathbf{y}_n + \mathbf{f}(t_n, \mathbf{y}_n) \Delta t + \frac{\mathbf{y}''(\xi)}{2} (\Delta t)^2, \quad \xi \in (t_n, t_{n+1})$$

The local truncation error is:

$$\tau_n = \frac{1}{\Delta t} (\mathbf{y}_{n+1} - \mathbf{z}_{n+1}) = \frac{\mathbf{y}''(\xi)}{2} \Delta t = \mathcal{O}(\Delta t).$$

Local truncation errors of common numerical methods for systems

Method	Local truncation errors τ_{n+1}
Forward Euler	$\mathcal{O}(\Delta t)$
Backward Euler	$\mathcal{O}(\Delta t)$
Trapezoidal	$\mathcal{O}((\Delta t)^2)$
Modified Euler	$\mathcal{O}((\Delta t)^2)$
Runge-Kutte (RK4)	$\mathcal{O}((\Delta t)^4)$

All common numerical methods are *convergent* with their *global truncation errors* $\mathbf{e}_{n+1}$ having the same order as the local truncation errors τ_{n+1} , when applied to a stable system and using a stable Δt .

Questions

- What is the largest Δt for which the Forward Euler method is stable around the point $(1, 0)$ for the system describing competing species (see Slide 22)?
- Derive the stability condition for the Modified Euler method applied to a stable test system with a diagonalizable matrix $A \in \mathbb{R}^{m \times m}$ with complex eigenvalues $\lambda_j, j = 1, \dots, m$.